

pharmaniaga

Adrenaline 1mg/mL Injection



COMPOSITION:

Each mL contains Adrenaline Acid Tartrate 1.8mg (equivalent to Adrenaline 1.0mg).

PRODUCT DESCRIPTION

A clear and colourless solution.

PHARMACODYNAMICS

Sympathomimetic amine. Non selective adrenoceptor stimulant. Adrenaline acts on the effector cells of sympathetic system. Some effector cells that respond to sympathetic nerve stimulation are stimulated by adrenaline, others are inhibited.

Adrenaline acts on both alpha and beta adrenergic receptors of tissue innervated by sympathetic nerves, except the sweat glands and arteries of the face. It is the most potent alpha receptor activator. Adrenaline stimulates the heart to increased output; raises the systolic blood pressure; lowers diastolic blood pressure; relaxes bronchial spasm and mobilises liver glycogen, resulting in hyperglycaemia and possibly glycosuria.

PHARMACOKINETICS

The effects after SC administration are produced within 5 minutes, which is slower than from IM administration. Endogenous plasma concentration in normal subjects are in the range 30 – 160 pg/mL.

Rapidly taken up by the heart, spleen, several granular tissues, and adrenergic nerves; only metabolites are detectable in CSF; readily crosses the placenta; protein binding, about 50% bound to plasma proteins.

Rapidly metabolised by oxidative deamination & O-methylation followed by reduction of glucuronic acid or sulphate conjugation.

73-95% of IV dose is excreted in the urine; of the material about 80% is excreted as O-methyl metabolites and 2% as catechol metabolites and only 1% as unchanged drug; the major urinary metabolite is 4 – hydroxy – 3 – methoxy – mandelic acid (HMMA); others include 4-hydroxy- 3-methoxyphenylacetic acid (HVA) conjugated metadrenaline, and 4 – hydroxy – 3 – methoxy-phenylglycol along with minor amounts of dihydroxymandelic acid in free or conjugated form and N-methyladrenaline.

INDICATION

It is used in the emergency treatment of anaphylactic shock due to reaction from insect bites and drugs and for symptomatic relief of serum sickness, urticaria, pruritus, angioneurotic oedema, respiratory distress due to bronchospasm, and relief of hypersensitivity reactions to drugs and other allergens.

RECOMMENDED DOSAGE

Usual adult and adolescent dose:

- Bronchodilator:- Subcutaneous, initially 200 to 500mcg (0.2 to 0.5mg) (base) repeated every 20 minutes to 4 hours as needed, the dosage being increased up to a maximum of 1mg per dose, if necessary.
- Anaphylactic reactions:- Intramuscular or subcutaneous, initially 200 to 500mcg (0.2 to 0.5mg) (base) repeated every 10 to 15 minutes as needed, the dosage being increased up to a maximum of 1mg per dose if necessary.
- Vasopressor (anaphylactic shock):- Intramuscular or subcutaneous, initially 500mcg (0.5mg) (base) repeated every 5 minutes if necessary; may be followed by intravenous administration of 25 to 50mcg (0.025 to 0.05mg) every 5 to 15 minutes as needed, if inadequate response to intramuscular or subcutaneous administration. Or:
- Intravenous, 100 to 250mcg (0.1 to 0.25mg) (base) administered slowly. May be repeated every 5 to 15 minutes as needed or followed by an intravenous infusion at an initial rate of 1mcg (0.001mg) per minute, the rate being increased to 4mcg (0.004mg) per minute, if necessary.

Usual paediatric dose:

- Brochodilator; or
- Anaphylactic reactions:- Subcutaneous, 10mcg (0.01mg) (base) per kg of body weight or 300mcg (0.3mg) per square meter of body surface up to a maximum of 500mcg (0.5mg) per dose, repeated

every 15 minutes for two doses, then every 4 hours as needed.

- Vasopressor (Anaphylactic Shock):- Intramuscular or subcutaneous, 10mcg (0.01mg) (base) per kg body weight, up to a maximum of 300mcg (0.3mg), repeated every 5 minutes if necessary. Intravenous, 10mcg (0.01mg) (base) per kg of body weight every 5 to 15 minutes as needed, if an inadequate response to intramuscular or subcutaneous administration.

ROUTE OF ADMINISTRATION

Parenteral

CONTRAINDICATIONS

Hyperthyroidism, hypertension or arteriosclerosis, narrow-angle glaucoma, angle-closure glaucoma (pre-operatively), hypersensitivity to sympathomimetic amines, shock (cardiogenic, traumatic or haemorrhagic), cardiac dilatation, coronary insufficiency, organic brain disease during general anaesthesia with halogenated hydrocarbons cyclopropane; or with local anaesthesia in fingers, toes, ears, nose or genitalia (there is a danger of vasoconstriction producing sloughing of tissues in these areas). It is also contraindicated during labour (it may delay the second stage by inhibiting spontaneous or oxytocin-induced contractions of the pregnant human uterus).

WARNINGS / PRECAUTIONS

Other beta-agonist sympathomimetics: Allow sufficient time to elapse before or after administering another beta-agonist sympathomimetic agent to avoid additive effects.

Disease states: Should be used cautiously in diabetic patients, elderly patients, cardiovascular disease patients, cerebrovascular insufficiency patient, hypertension, chronic lung disease, angina pectoris, hyperthyroidism, bronchial asthma, psychoneuroses, tricyclic antidepressants, some antihistamines and thyroid hormones.

General anaesthesia: Concurrent use with cyclopropane or halogenated hydrocarbon anaesthetics may produce fatal ventricular arrhythmias.

Diabetic patients: A greater increase may be produced in heart rate, blood glucose, lactate, glycerol and free fatty acids when adrenaline is administered to diabetic patients with autonomic neuropathy than in diabetic with neuropathy.

Sodium Metabisulphite: This product contains Sodium Metabisulphite that may cause serious allergic type reactions in certain susceptible patients. Do not use if known to be hypersensitive to bisulphites.

The possibility of an allergic reaction to Sodium Metabisulphite should be considered in asthmatic patients who show paradoxical worsening of their condition following use of the drug.

Gangrene: Inter-arterial administration must be avoided as marked vasoconstriction may result in gangrene.

Should not be used in presence of ventricular hyperexcitability produced by chloroform, halothane, digitalis, mercurial diuretics or quinidine.

INTERACTION WITH OTHER MEDICAMENTS

Other sympathomimetic agents: Adrenaline should not be administered concomitantly with other sympathomimetic agents because of the possibility of additive effects and increased toxicity.

General anaesthesia: Administration of adrenaline in patients receiving cyclopropane or halogenated hydrocarbon general anaesthetics that increase cardiac irritability and seem to sensitise the myocardium to adrenaline, may result in arrhythmias including ventricular premature contractions, tachycardia or fibrillation.

Cardiovascular drugs: Adrenaline should not be used in patients receiving high dosage of other drugs, eg. Digoxin, that can sensitise the heart to arrhythmias.

CNS and other drugs: Tricyclic antidepressants, some antidepressants, some antihistamines and thyroid hormones may potentiate the effects of adrenaline, especially on heart rhythm and rate.

Beta Blockers: The administration of adrenaline to patients receiving non-selective beta blockers (eg. Propranolol) may result in severe hypotension, followed by a reflex bradycardia, due to stimulation of adrenergic receptors.

Cautions also needed with α -adrenergic blocking agents, parenteral-local anaesthetics, oral antidiabetic agents, insulin diatrizoates, Iothalamate, Ioxaglate, Ergot alkaloids, Methysergide, Oxytocin, Doxapram, Guanadrel, Guanethidine, Mazindol, Mecamylamine, Methyldopa, Trimethaphan, Methylphenidate and Rauwolfia alkaloids.

Incompatibilities

Adrenaline is rapidly denatured by oxidising agents and

alkalis including sodium bicarbonate, halogens, nitrates, nitrites and salts of iron, copper and zinc. Adrenaline may be mixed with 0.9% Sodium Chloride injection but is incompatible with 5% sodium chloride injection. The stability of Adrenaline in 5% dextrose injection decreases when the pH is greater than 5.5.

PREGNANCY AND LACTATION

Pregnancy

Adrenaline has been administered to a large number of pregnant women and women of childbearing age without any proven increase in the frequency of malformations or other direct or indirect harmful effects on the foetus having been observed.

However, the use of adrenaline during labour is contra-indicated because it may delay the second stage by inhibiting spontaneous or oxytocin-induced contractions of the pregnant human uterus.

Lactation

Adrenaline is excreted in the breast milk. The use of adrenaline in breast-feeding women is therefore not recommended.

SIDE EFFECTS

- Anxiety, dyspnoea, hyperglycaemia, restlessness, palpitation, tachycardia, tremors, weakness, dizziness, headache, vomiting and coldness of extremities may occur even with small doses of adrenaline.
- Hypertension, high blood pressure may produce cerebral haemorrhage and pulmonary oedema.
- Others include difficulty in micturation and urinary retention, weakness, altered metabolism including disturbance of glucose metabolism, sweating and hypertension.

Adrenaline may cause reactions such as fear, tenseness, pallor and respiratory difficulty. Cerebrovascular and hemiplegia may result, especially in geriatric patients. Inadvertent IV injection of adrenaline has also been reported to have caused convulsions, metabolic acidosis, and renal failure with anuria. Repeated injections of adrenaline can cause necrosis as a result of vascular constriction at the injection site. Prolonged use or overdosage of adrenaline can result in severe metabolic acidosis. Gas gangrene which can be fatal has been reported following intramuscular injection of adrenaline into the buttock or thigh. This appears to have been due to Clostridium organism on the skin being deposited into muscle tissue during injection, with the vasoconstrictor properties of adrenaline enhancing

the effects of the injection. High doses may result in ventricular arrhythmias. Rigidity and tremor may be exacerbated in patients with Parkinsonism Syncopal episodes have been reported in children Psychiatric disorders may be exacerbated.

SYMPTOM & TREATMENT OF OVERDOSE

Symptoms: Initial hypertensive episode followed by hypotension. Cardiac arrhythmias, cerebral haemorrhage and pulmonary oedema.

Treatments: Rapid digitalisation with digoxin 1mg intravenously in divided doses. Or immediate IV injection of quick acting alpha-adrenoceptor blocking agent (eg. 5 – 10mg Phentolamine Mesylate) followed by a beta-adrenoceptor blocking agent (eg. 2.5 to 5mg propranolol). Rapidly acting vasodilators such as Glyceryl Trinitrate has also been used.

Effects on ability to drive and use of machines

Adrenaline has moderate influence on the ability to drive and use machines. The patients' ability to drive and use machines may be affected by the anaphylactic reaction, as well as by possible adverse reactions to adrenaline.

Instructions for Use

Please refer to the Recommended Dosage section.

For S.C. or I.M. Injection.

Use as directed by the physician.

Keep out of sight and reach of children.

If only part used, discard the remaining solution.

Any unused medicinal product or waste material should be disposed of in accordance with local requirement.

STORAGE CONDITION

Store below 30°C. Protect from light.

Caution: Do not use if solution contains particles, growth, turbidity or if there is any change of appearance.

SHELF LIFE

2 years after date of manufacture.

DOSAGE FORMS AND PACKAGING AVAILABLE

- 10 x 1mL ampoule (amber colour)

PRODUCT REGISTRATION HOLDER

/MANUFACTURER

Pharmaniaga Lifescience Sdn Bhd (82685-T)

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PM216.1

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